Pythia training dynamics by update range

Pythia-14M Pythia-31M Pythia-70M

Updates 1-100 | N=5

Updates 100-400 | N=21

Change in ln(training loss)

$\hat{\eta}\hat{\kappa}_t$ (proxy)

Raw adjacent cosine ρ_t

Raw gradient $\|g_t\|_2$

Optimizer update

Optimizer update

All rows: complete trailing means, N=5 left and N=21 right. Loss uses $\ln L_t - \ln L_{t-1}$. Seed 1235.

Y limits vary by column; norm row uses log Y. Update 100 appears in both columns; smoothing spans the boundary.

Left means start at update 5 (norm) / 6 (other rows). Update 100 has different smoothing windows in the two columns.